

BUILDING DETECTION FROM SATELLITE IMAGERY

YOLOv8 Based Object Detection Project Report

1. Abstract

This project presents an automated system for detecting and counting buildings in satellite imagery using the YOLOv8 object detection model. The system processes high-resolution satellite images and identifies building structures using bounding box predictions. The model is trained on the SpaceNet 2 Paris dataset and achieves strong performance in both detection accuracy and counting reliability.

2. Introduction

Manual analysis of satellite images is time-consuming and inefficient. With the advancement of deep learning, object detection models like YOLOv8 enable automated detection of objects such as buildings at scale.

The objective of this project is to:

- Detect buildings in satellite images
- Count the number of buildings automatically
- Evaluate model performance using standard metrics

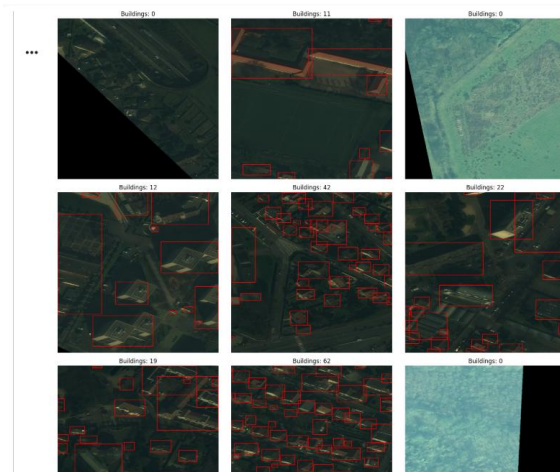
3. Dataset Description

The dataset used is the SpaceNet 2 – Paris Buildings Dataset, which contains satellite images annotated with building footprints.

Dataset Statistics:

- Total images: 1148
- Image resolution: 650 × 650 pixels
- Total building annotations: 16,478
- Average buildings per image: 14.35

Each image has a corresponding YOLO-format label file containing bounding box coordinates.



4. Data Preprocessing

Before training, the dataset was prepared as follows:

- Converted GeoTIFF images into PNG format for YOLO compatibility
- Verified image integrity and resolution consistency
- Split dataset into:
 - Training set: 918 images
 - Validation set: 115 images
 - Test set: 115 images
- Converted annotations into YOLO format (normalized bounding boxes)

```
TRAIN
Images: 918
Labels: 918
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```
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VAL
Images: 115
Labels: 115
```

```
-----
TEST
Images: 115
Labels: 115
-----
```

5. Exploratory Data Analysis

Initial analysis of the dataset revealed:

- All images are consistent in size (650 × 650)
- Building density varies significantly across images
- Some images contain dense urban clusters
- Some images contain few or no buildings

Challenges Identified:

- Small rooftops difficult to detect

- Overlapping buildings in dense areas
- Variation in building shapes and sizes
- Background noise (roads, vegetation, shadows)



Sparse building region



Dense building region

6. Model Architecture

The YOLOv8n (nano version) model was used for this project.

Why YOLOv8?

- Fast real-time object detection
- Lightweight and efficient
- Pretrained on large-scale datasets
- Good balance between speed and accuracy

Pretrained weights (yolov8n.pt) were used to improve convergence and reduce training time.

7. Training Details

- Epochs: 20
- Image size: 640x640
- Batch size: 16
- Device: GPU (Tesla T4 on Google Colab)
- Framework: Ultralytics YOLOv8

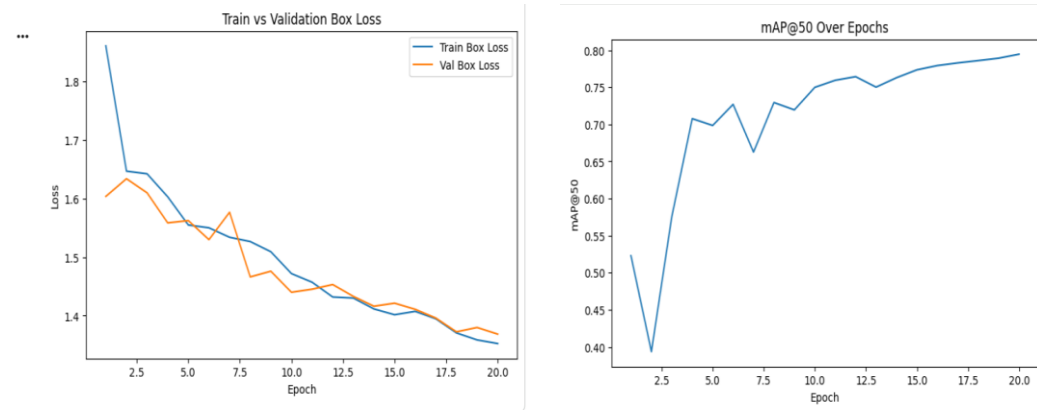
8. Training Results

The model was evaluated using validation data during training.

Final Metrics:

- Precision: 0.830
- Recall: 0.707

- mAP@50: 0.795
- mAP@50–95: 0.468
- F1 Score: 0.764
- MAE (Counting Error): 0.6



9. Model Evaluation

The model performance was further evaluated using:

- Precision-Recall Curve
- Confusion Matrix
- Test set predictions

10. Building Counting System

A custom pipeline was developed to:

- Detect buildings in satellite images
- Count total buildings per image
- Compute average prediction confidence
- Measure processing time per image

Example Output:

- Image: test_042.png
- Buildings detected: 7
- Average confidence: 0.73
- Processing time: 84 ms

11. Results Summary

Metric	Value
Precision	0.830
Recall	0.707
mAP@50	0.795
mAP@50–95	0.468
F1 Score	0.764
MAE	0.6

12. Conclusion

This project successfully demonstrates the use of YOLOv8 for automated building detection in satellite imagery. The model achieved strong detection accuracy and low counting error, making it suitable for geospatial analysis tasks.

13. Future Improvements

- Train for more epochs (50-100)
- Use YOLOv8sd or YOLOv8m for higher accuracy
- Improve detection in dense urban regions
- Apply post-processing for better counting accuracy